Machine Learning in Azure Cloud

Azure Cloud

VM (Virtual Machine)

Machine Learning Studio

Synapse Analytics

Running models in SQL

Power BI

VM or DSVM

VM (Virtual Machine) or **DSVM** (Data Science VM)
They are the same price, but DSVM comes with pre-installed software
(python 3.6.9, sklearn, tensorflow, azureml, etc.)

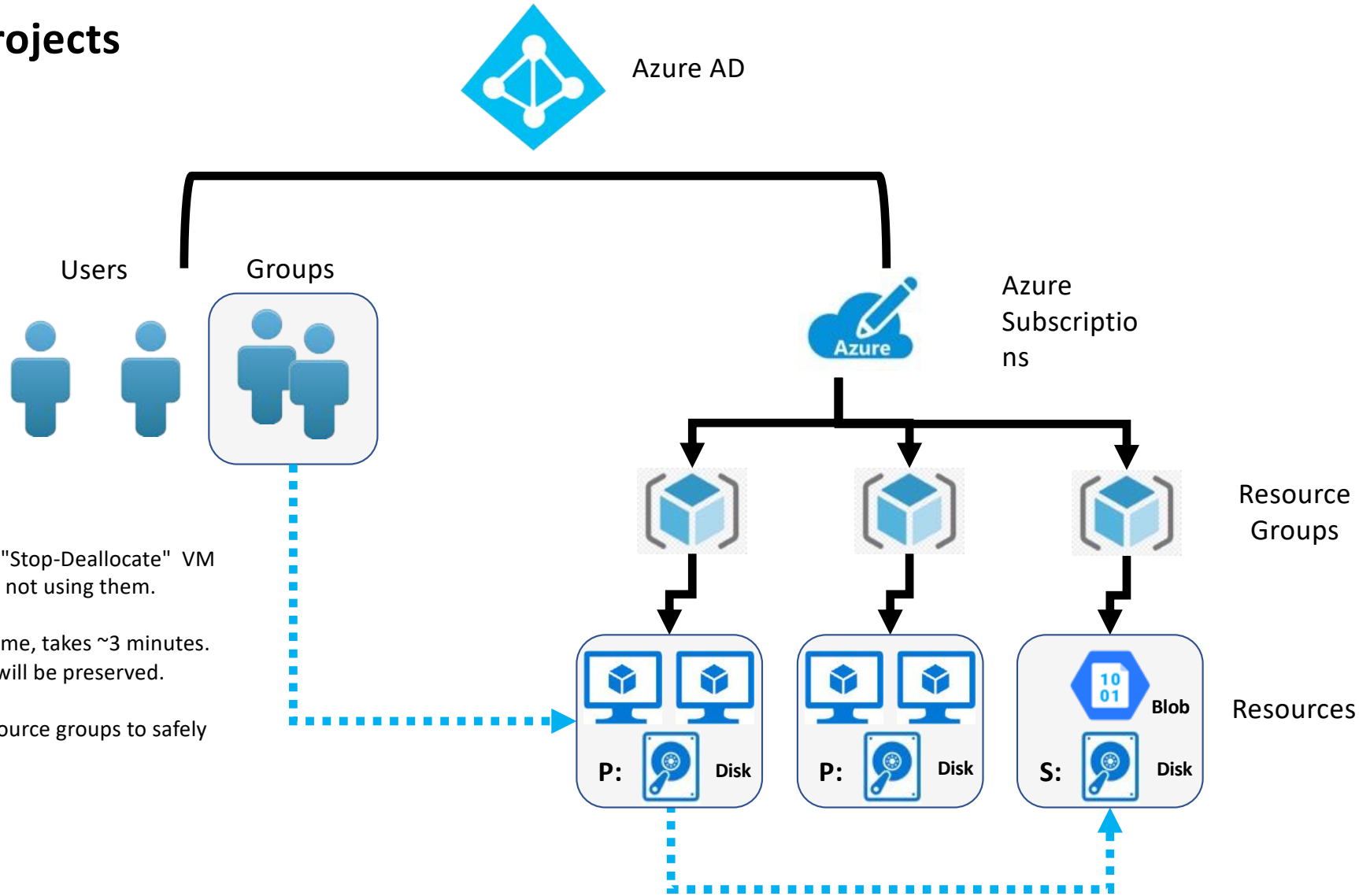
- You get admin rights, can install any tools you need
- Can install conda python and any ML libraries you want
- Can connect via ssh, can use GitHub
- Can work remotely using Jupyter notebooks
- Can use local editors with built-in sftp transfer
- (BBEdit on Mac, UltraEdit on Windows, etc.)
- Can use Azure CLI (Command Line Interface) to manage azure resources and data.
- Can use azure python libraries (to move data around, etc.)

Linux VM Pricing
https://azure.microsoft.com/en-us/pricing/details/virtual-machines/linux/
Azure VM Pricing Comparison
https://azureprice.net/?region=eastus2&cores=16,16&ram=0,11400
Azure ML Pricing
https://azure.microsoft.com/en-us/pricing/details/machine-learning/
Azure Synapse Analytics Pricing
https://azure.microsoft.com/en-us/pricing/details/synapse-analytics/
Managed Disk Pricing
https://azure.microsoft.com/en-us/pricing/details/managed-disks/

Lowest prices are in regions East US 2 or West US 2
Examples of prices for Linux VMs without discounts, "pay-as-you-go":

VM Type	Cores	RAM GB	swap GB	OS disk GB	GPU	$/month	$ for extra 1TB fast SSD
Ddv2	2	7	100	14	-	101	123
Ddv4	16	64	600	30	-	650	123
NC12 Promo	12	112	680	30	2xK80	600	123
NC12	12	112	680	30	2xK80	1,336	123

Separate Projects



To save money you can "Stop-Deallocate" VM instances when you are not using them.

You can restart at any time, takes ~3 minutes. All disks (except swap) will be preserved.

You can use AD and resource groups to safely separate projects.

Azure Machine Learning

Machine Learning Workspace – set of tools for ML

Azure Machine Learning Studio:
- Linux VMs (single or clusters)
- ML Studio notebooks
- Jupyter notebooks, or Jupyter Lab, or R studio
- python scripts (or R) – can run in terminal
- data storage access
- AutoML (regression, classification, forecasting)
- automating experiments and whole ML workflows (manually or from scripts):
  .. create datasets (data containers)
  .. models (model containers)
  .. pipelines (executable flow)
  .. endpoints (web or for batch)
  .. experiments - submitted pipeline

Note:

"**Classic**" is the older version of Azure ML Studio
where there is no Jupyter notebook.
You can only drag & drop pieces.
It is basically the "Designer" part of the new ML Studio.

Azure ML : https://ml.azure.com

- Login into https://portal.azure.com
- Move mouse to top-right and click on your icon (sign-in/sign-out) and select a directory (you may have several).
- Then select subscription
- Then go to Machine Learning workspace (can search for it, or it is in the list, you may have several ML workspaces)
  - https://ml.azure.com/selectWorkspace
- One in ML workspace, find and click the url to **Azure Machine Learning Studio.**
- Here you can work with python notebooks, etc.

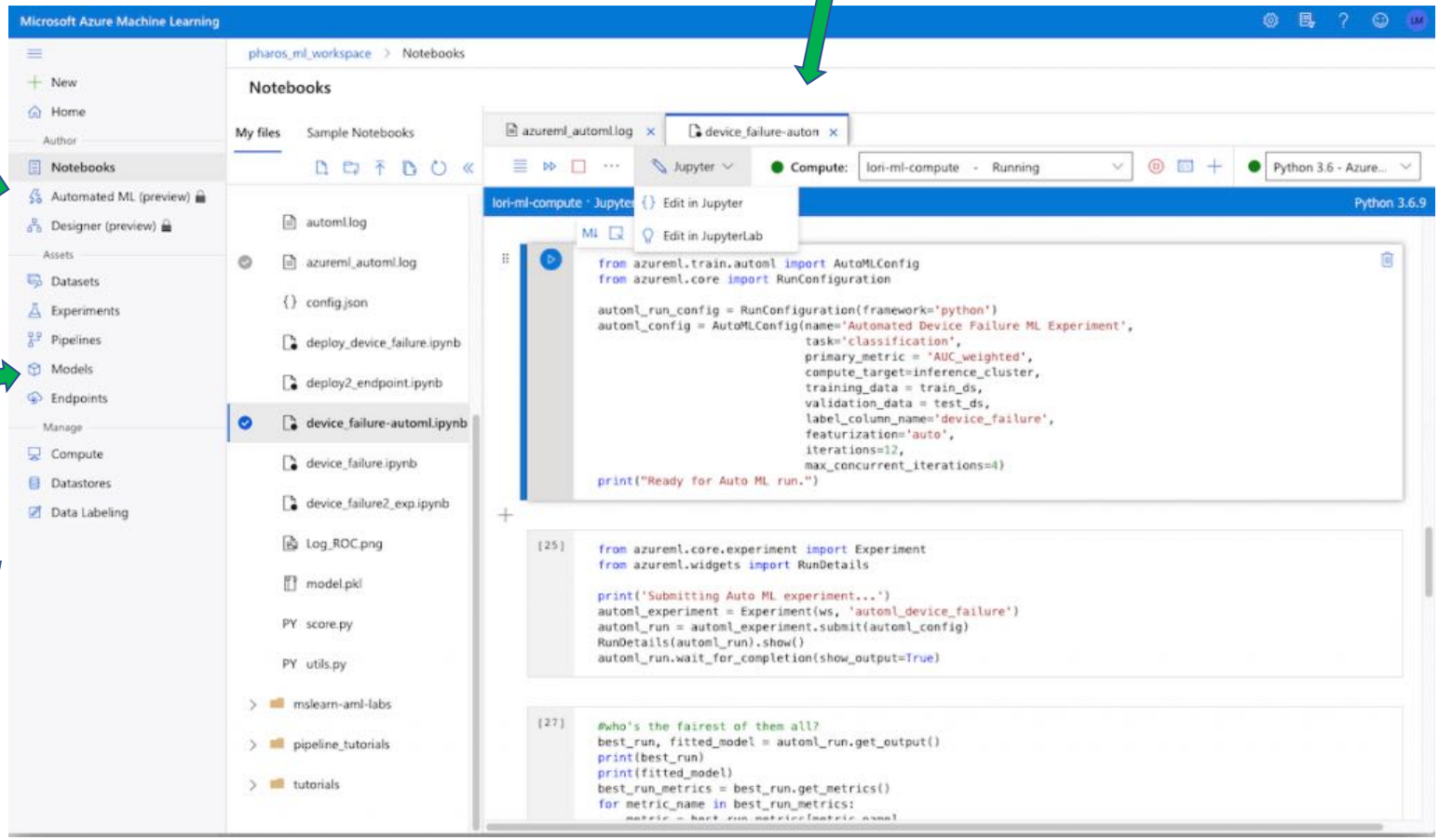
Machine Learning Studio

Multiple tabs with notebooks and scripts

Author:
- Notebooks
- AutoML
- Designer

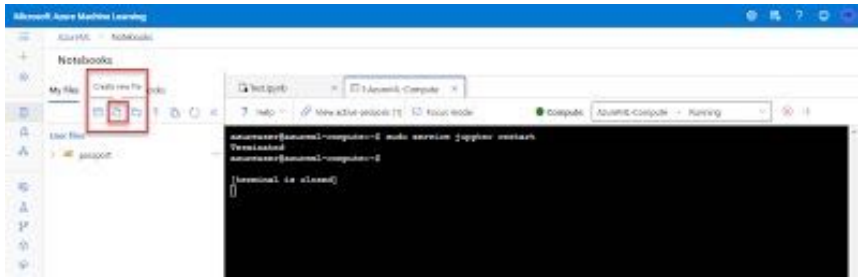
Assets:
- Datasets
- Experiments
- Pipelines
- Models
- Endpoints

Manage:
- Compute
- Datastores
- Data Labeling



Machine Learning Studio - continued

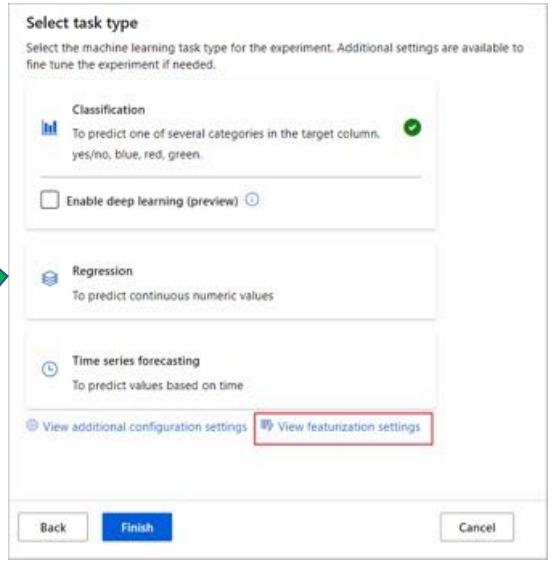
Running scripts from unix terminal



Selecting AutoML Type

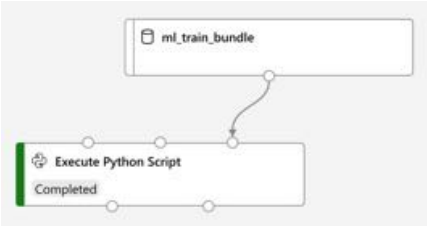
AutoML can run ~200 models and select the best one for you.

You create AutoML by going through a set of screens
where you can select the type of the model
(Classification, Regression, Time series forecasting), selecting the
data (features and target),
types of features (categorical, ...),
narrow down types of models, etc.



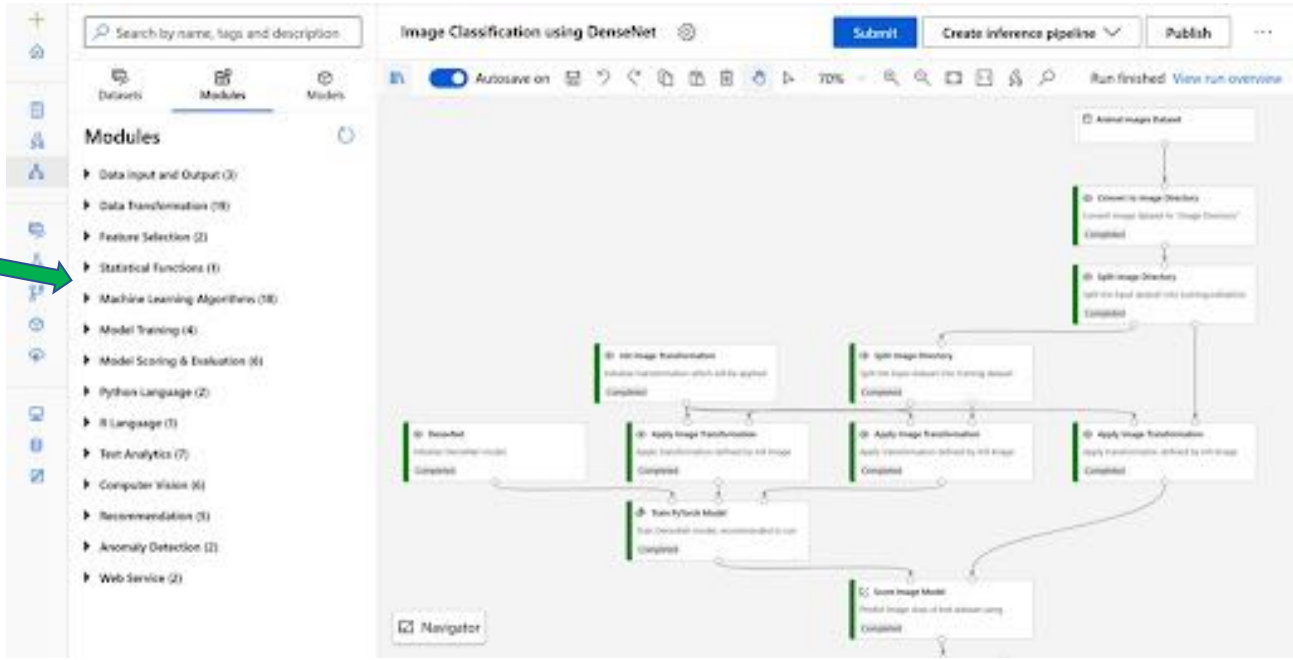
Machine Learning Studio - continued

Simple pipeline to run python script with custom modules



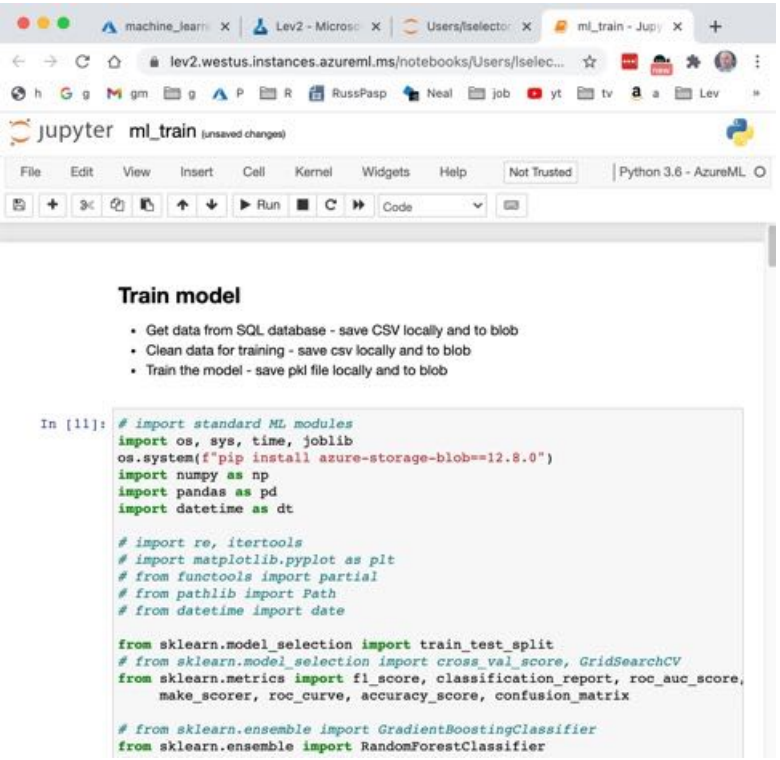
Datasets
Sample datasets
Data Input and Output
Data Transformation
Feature Selection
Statistical Functions
Machine Learning Algorithms
Model Training
Model Scoring & Evaluation
Python Language
R Language
Text Analytics
Computer Vision
Recommendation
Anomaly Detection
Web Service

Complex pipeline

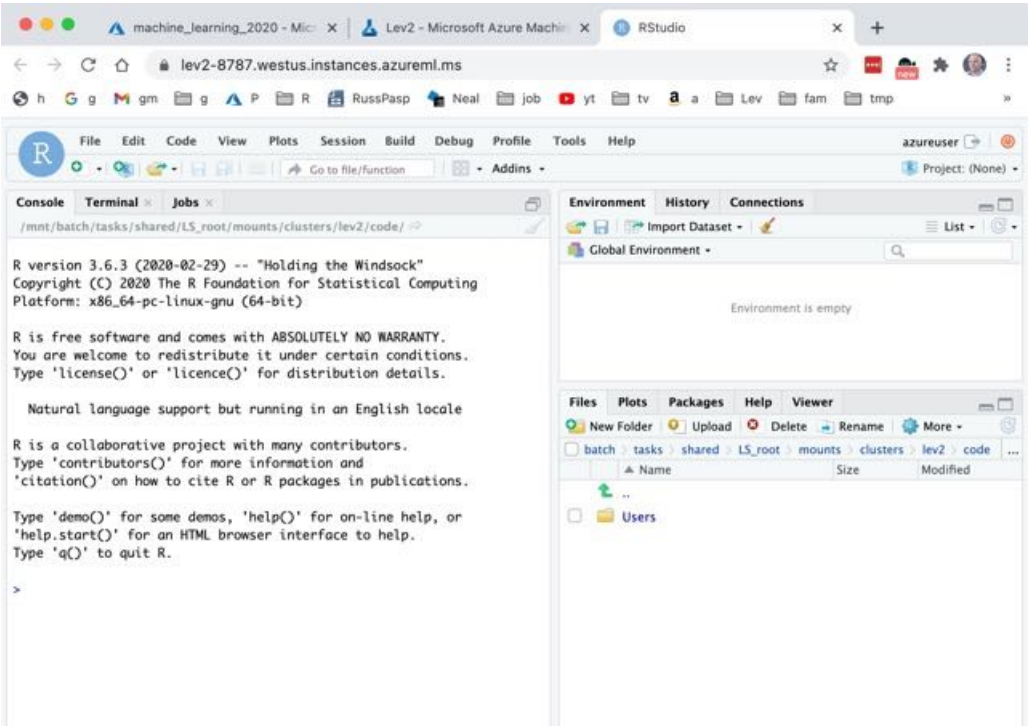


Machine Learning Studio - continued

Jupyter Notebook or Jupyter Lab

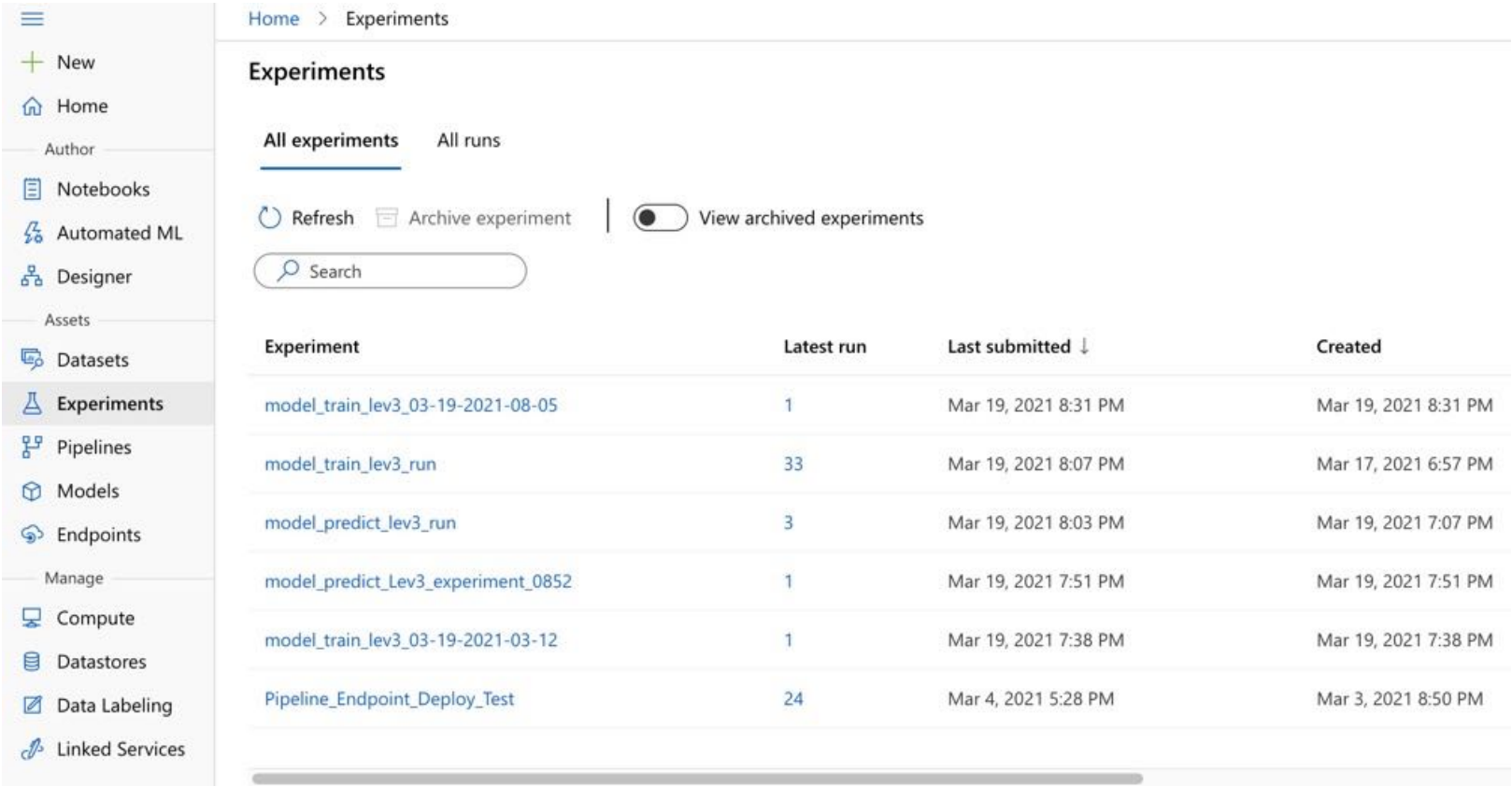


R-Studio



Machine Learning Studio - continued

Running Experiments



Machine Learning Studio - authentication

To run ML jobs you need to authenticate
using one of three main methods:

- **Interactive** – per-user authentication during experimentation and iterative development. Can prompt user to authenticate.
- **Service principal** – non-human account to run your jobs automatically. Need name/password (called "clientId" and "clientSecret") - https://docs.microsoft.com/en-us/azure/machine-learning/how-to-setup-authentication#configure-a-service-principal
- **Managed identity** – used when using ML SDK on Azure VM or cluster. Allows to connect to workspace without storing credentials in Python code or prompting the user to authenticate

Example: Service principal authentication in python script:

```
from azureml.core.authentication import InteractiveLoginAuthentication
from azureml.core.workspace import Workspace
from azureml.core.authentication import ServicePrincipalAuthentication

# interactive_auth = InteractiveLoginAuthentication(tenant_id = "your-ad-tenant-id")

sp = ServicePrincipalAuthentication(
    tenant_id="your-ad-tenant-id",
    service_principal_id="clientId",
    service_principal_password="clientSecret"
)

# load existing worksapce
bag.ws = Workspace.get(name='your-ml-workspace-name',
               auth=sp,
               subscription_id='your-subscription-name',
               resource_group='your-resource-group-name'
)
```

Azure ML : Some Links

Azure ML Pricing - https://azure.microsoft.com/en-us/pricing/details/machine-learning/

Here is a very good post on training using sklearn – and then register the model in ML Studio:
https://towardsdatascience.com/deploying-ml-models-on-azure-a948c106f7b5
or https://ai.plainenglish.io/deploying-ml-models-on-azure-8daa9c10db5a

This above tutorial doesn't actually "deploy" the model.
The more appropriate title should be "**Register your custom python model in ML Studio**"
Note the use of library "**joblib**" to store the model into pickle (and to retrieve it).

```
from sklearn.externals import joblib
joblib.dump(value=mymodel, filename="mymodel.pkl")
```

and then:

```
from azureml.core.model import Model
model = Model.register(
    workspace=ws,
    model_path="mymodel.pkl",
    model_name="boston_housing_lr")
```

Then you can use the second tutorial on how to deploy the model:
https://docs.microsoft.com/en-us/azure/machine-learning/how-to-deploy-and-where

For this deployment you will need to write the "**entry script**" with methods **init() and run()**
Once you have this script, you can go to your models (in ML Studio),
click on your model, then click on "Deploy" on top,
fill-in parameters (name, ... , entry script, ...) - and click the button to deploy it.

Another (simpler) way to deploy the model is to create a ML pipeline with the python script,
and then execute it from ADF (Azure Data Factory):
https://docs.microsoft.com/en-us/azure/data-factory/transform-data-machine-learning-service

Very good youtube video "**Train and deploy ML models at scale using Azure Machine Learning | INT134C**" -
- https://www.youtube.com/watch?v=2OgBxYCnVGk

And here is the link from this video to a good 10-hrs course:
"**Build AI solutions with Azure Machine Learning**"
- https://docs.microsoft.com/en-us/learn/paths/build-ai-solutions-with-azure-ml-service/

Yet another good tutorial:
"**Creating End-to-End MLOps pipelines**"
- https://benalexkeen.com/creating-end-to-end-mlops-pipelines-using-azure-ml-and-azure-pipelines-part-1/

Machine Learning Workshop:
- https://github.com/microsoft/MCW-Machine-Learning
ML Ops Workshop:
- https://github.com/microsoft/MCW-ML-Ops

Also:
- https://docs.microsoft.com/en-us/azure/machine-learning/
- https://docs.microsoft.com/en-us/azure/machine-learning/tutorial-pipeline-batch-scoring-classification

- https://azure.microsoft.com/en-us/blog/retraining-and-updating-azure-machine-learning-models-with-azure-data-factory/
- https://azure.microsoft.com/en-us/blog/azure-data-factory-updates-integration-with-azure-machine-learning-2/
- https://docs.microsoft.com/en-us/azure///data-factory/transform-data-machine-learning-service
- Pipelines - https://www.youtube.com/watch?v=lVP7ywRq8p4
- Old (2016) video about calling ML from Azure Data Factory - https://www.youtube.com/watch?v=7k4XvM4wd0k

Azure ML : More Links

Google-ing mount datastore azure ml notebook -
- https://github.com/microsoft/MLOpsPython -
- https://docs.microsoft.com/en-us/python/api/azureml-core/azureml.core.datastore?view=azure-ml-py -

Develop for Azure Files with Python -
- https://docs.microsoft.com/en-us/azure/storage/files/storage-python-how-to-use-file-storage -

Azure for Python Developers -
- https://docs.microsoft.com/en-us/azure/developer/python/ -

Azure REST API Reference -
- https://docs.microsoft.com/en-us/rest/api/azure/ -

Azure SDK for Python -
- https://github.com/Azure/azure-sdk-for-python/tree/master/sdk/storage -
- https://docs.microsoft.com/en-us/python/api/overview/azure/ml/?view=azure-ml-py -

Azure SDK – trigger published pipeline -
- https://docs.microsoft.com/en-us/azure/machine-learning/how-to-trigger-published-pipeline#trigger-pipelines-with-azure-machine-learning-sdk-for-python –

Automate ML Experiments
- https://docs.microsoft.com/en-us/azure/machine-learning/how-to-use-automlstep-in-pipelines -
- https://towardsdatascience.com/hidden-tricks-for-running-automl-experiment-from-azure-machine-learning-sdk-915d4e3f840e -

Understand AutoML
- https://docs.microsoft.com/en-us/azure/machine-learning/how-to-understand-automated-ml#regressionforecasting-metrics -
- https://docs.microsoft.com/en-us/azure/machine-learning/how-to-configure-auto-train#primary-metric -

What is Automated ML
- https://docs.microsoft.com/en-us/azure/machine-learning/concept-automated-ml -

Many Models Solution Accelerator
- https://github.com/microsoft/solution-accelerator-many-models -

Configure automated ML experiments in Python:
- https://docs.microsoft.com/en-us/azure/machine-learning/how-to-configure-auto-train -

Azure Synapse Analytics

Azure Synapse Workspace – newest collection of Azure tools for Analytics - https://azure.microsoft.com/en-us/services/synapse-analytics/

- **Azure Storage Gen 2**
  substitutes for Storage Gen1 and Data Lake Gen 1.
  Gen 2 is more reliable, efficient, cheaper, supports Big Data

- **Data pools** (Serverless, Dedicated, Spark).
  Dedicated SQL Pool = SQL DataWarehouse
  You can query csv, jason, parquet files - serverless SQL pools

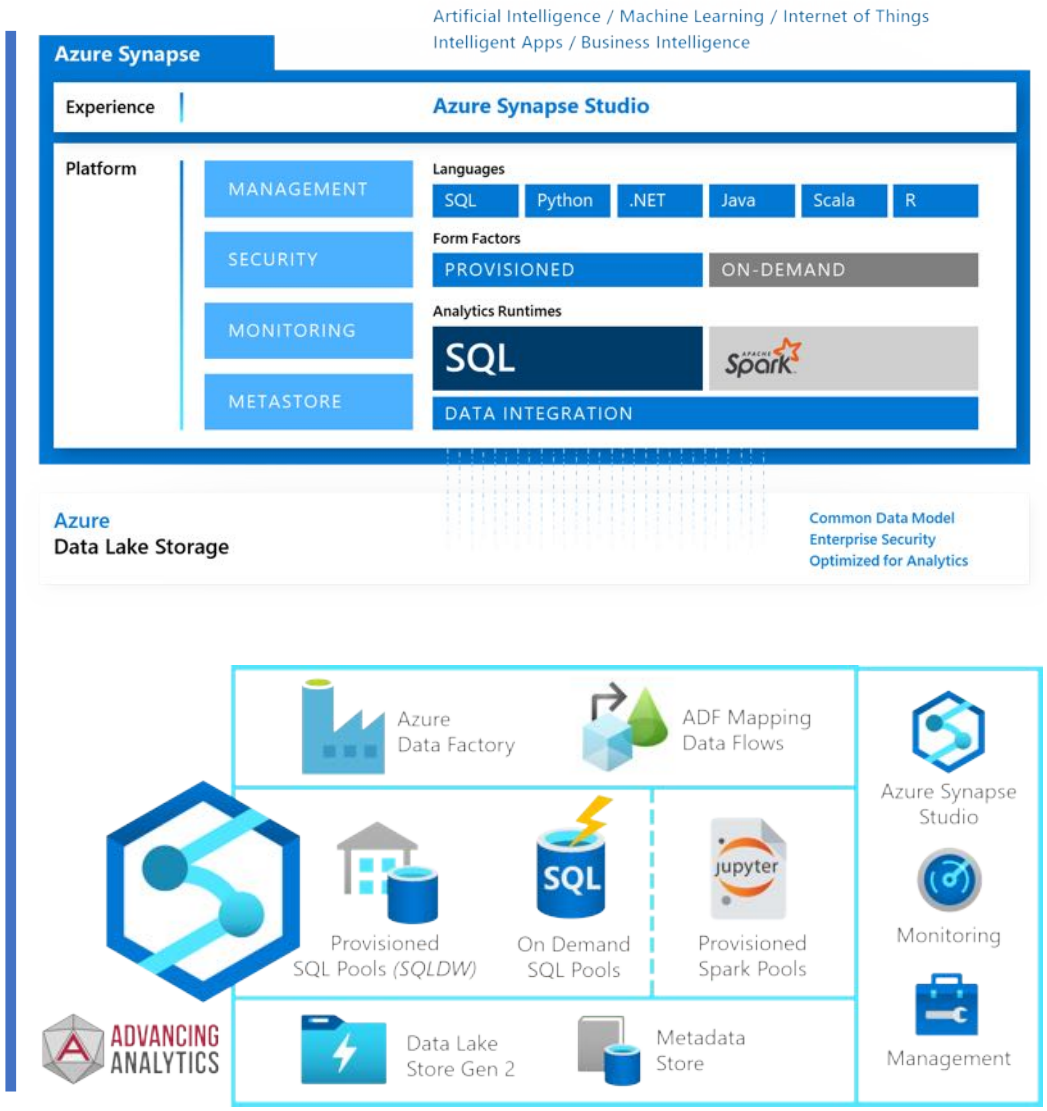
- **Compute clusters** – including Spark

- **Data Factory** – to automate data flows

- **Azure Synapse Link for Azure Cosmos DB**
  https://docs.microsoft.com/en-us/azure/cosmos-db/synapse-link

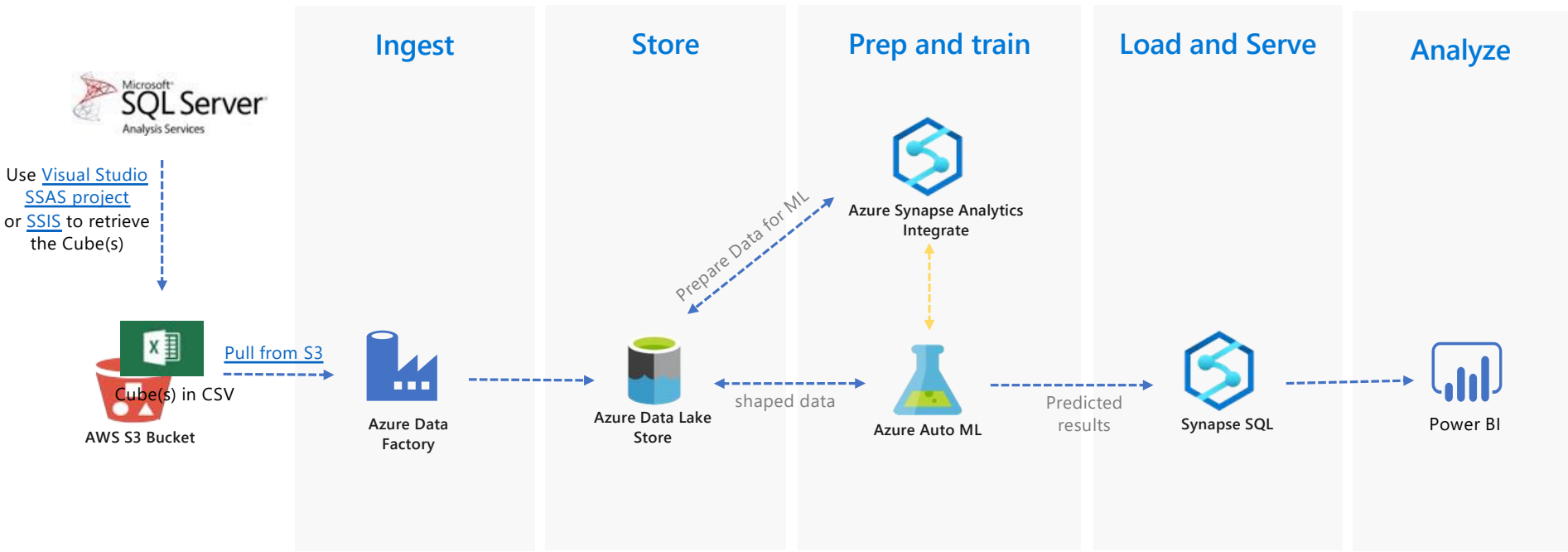
- **Cosmos DB Analytical Store**
  https://docs.microsoft.com/en-us/azure/cosmos-db/analytical-store-introduction

- **Power BI** Integration



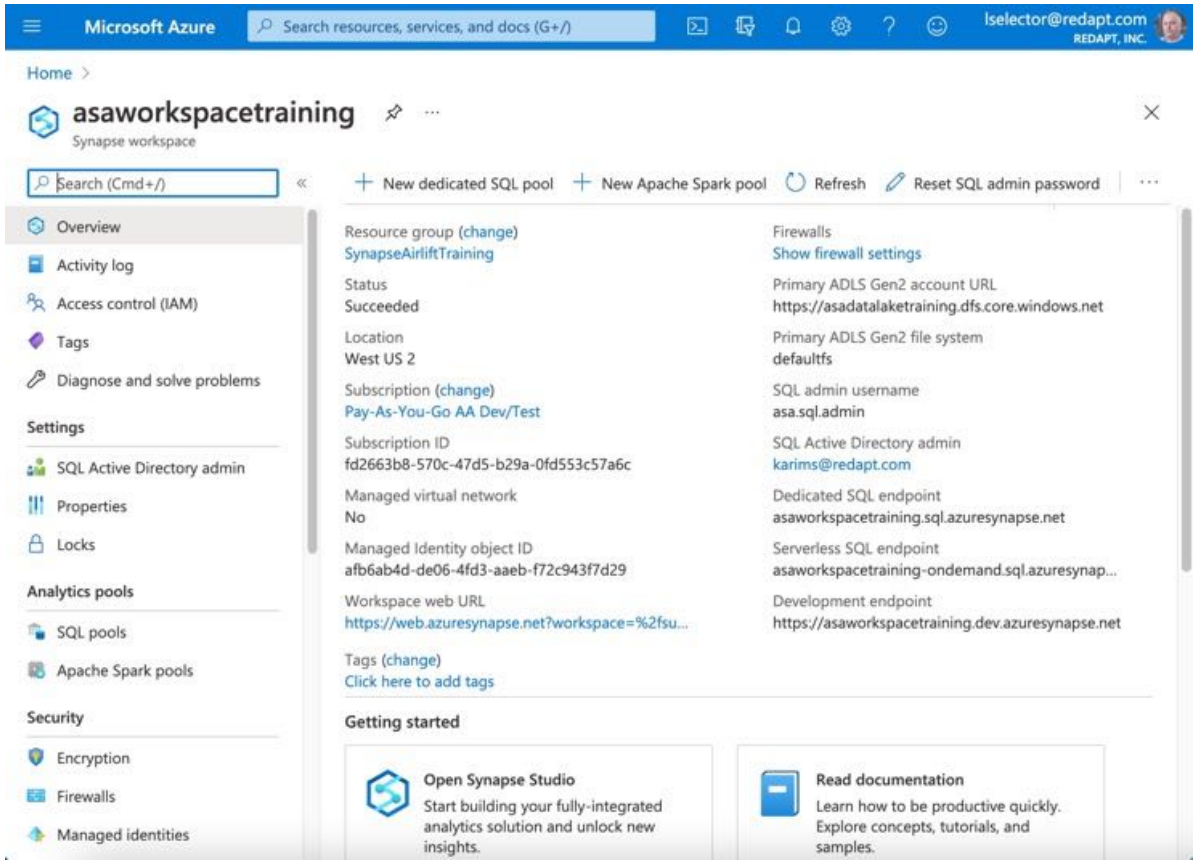
Common ML Architecture

(migrating data from AWS)



Create Synapse Analytics Workspace

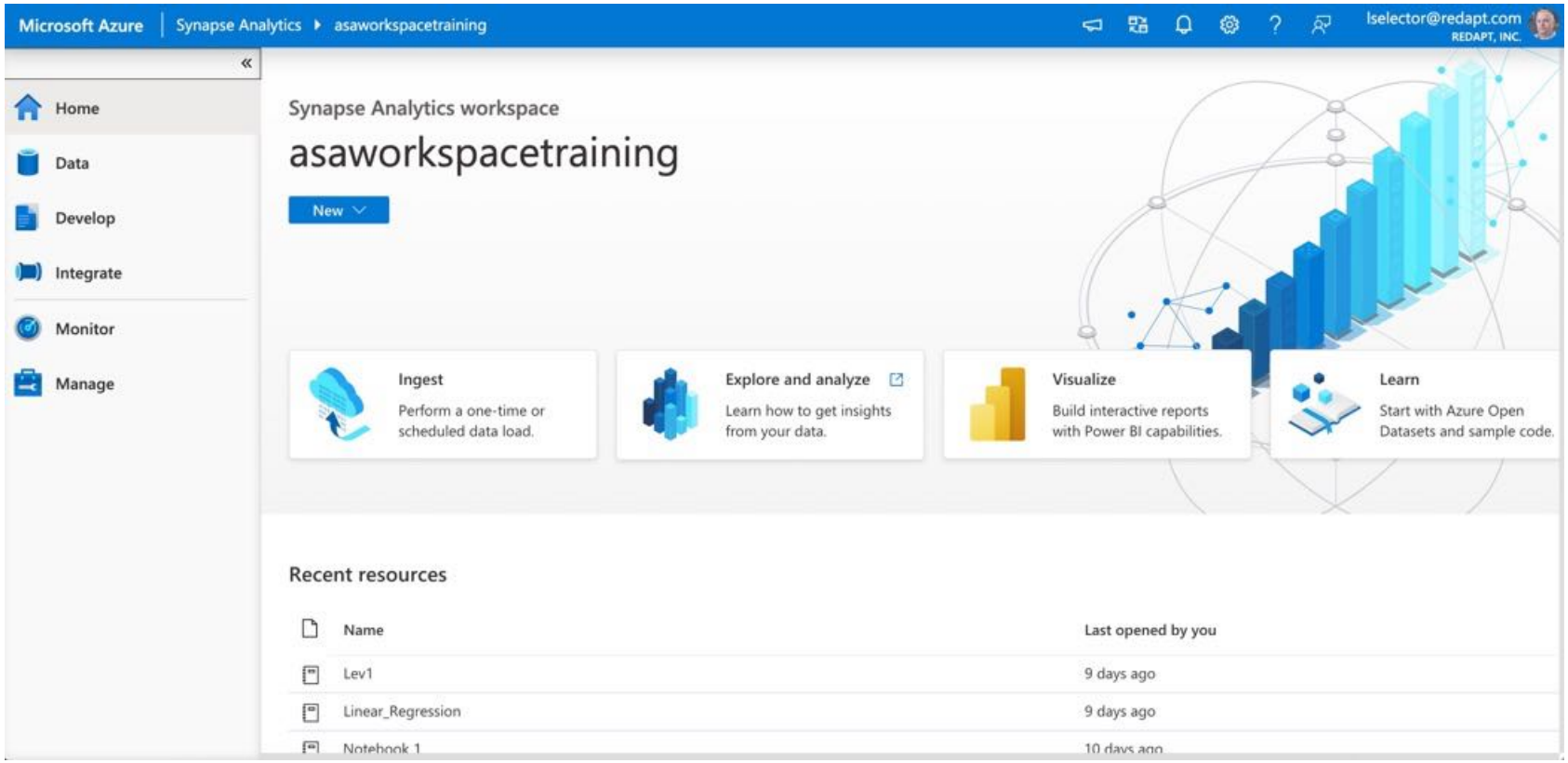
Then click on "Open Synapse Studio



Synapse Analytics Studio

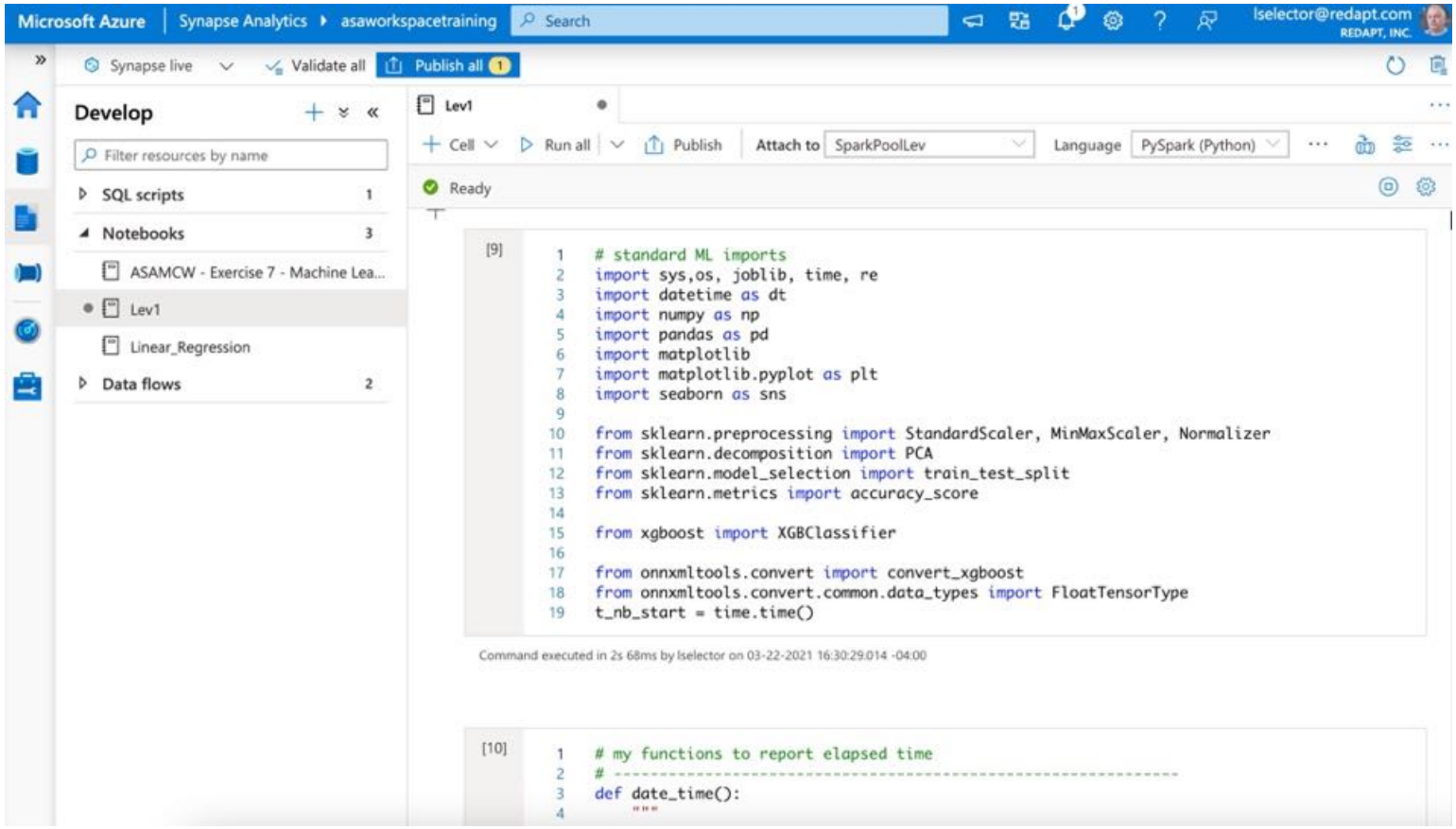
SQL script
Notebook
Data flow
Apache Spark job

pipeline
copy data



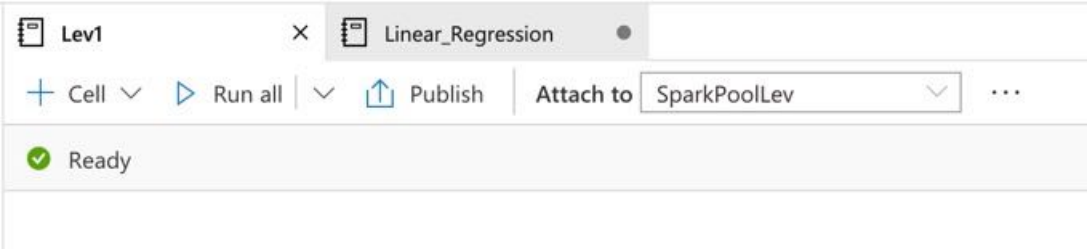
Synapse Notebook

You can run different languages – Python, SQL, Scala, etc.
Over a compute cluster – easy to resize as needed.
(shuts down after some idle time you specify).

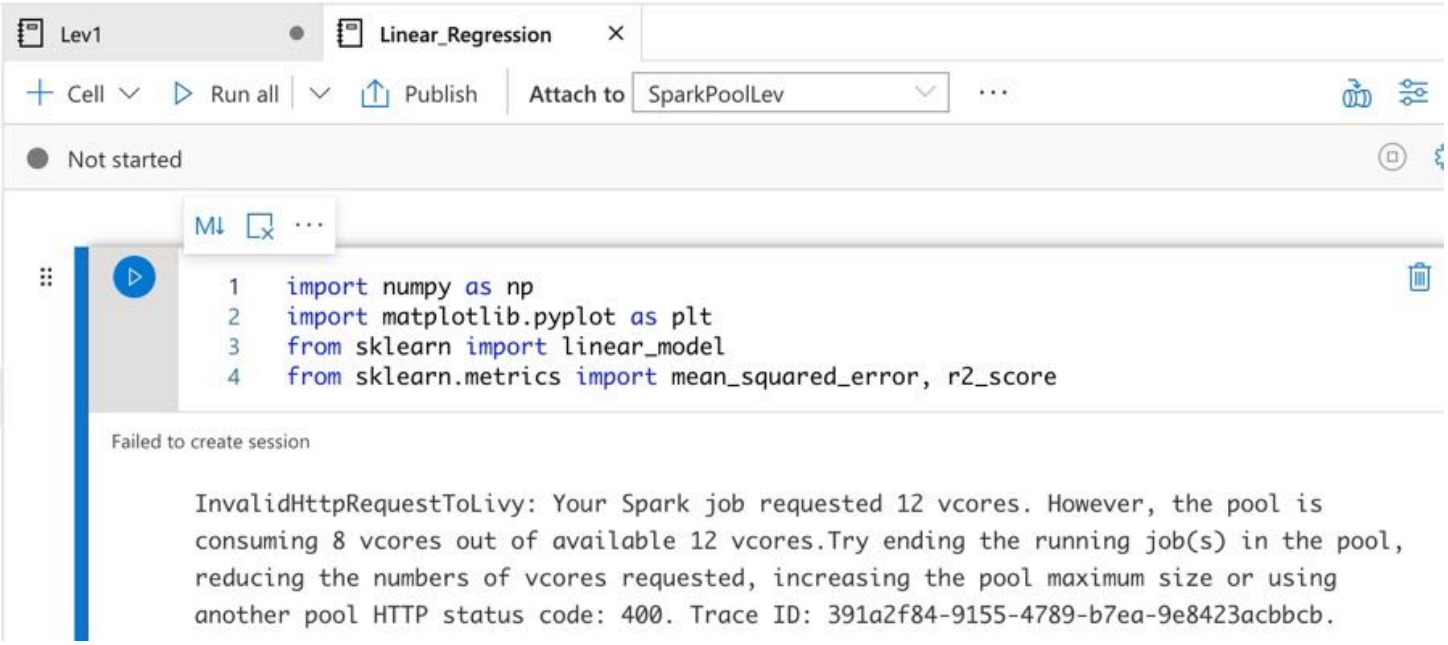


Each Notebook Gets Its Own Cluster

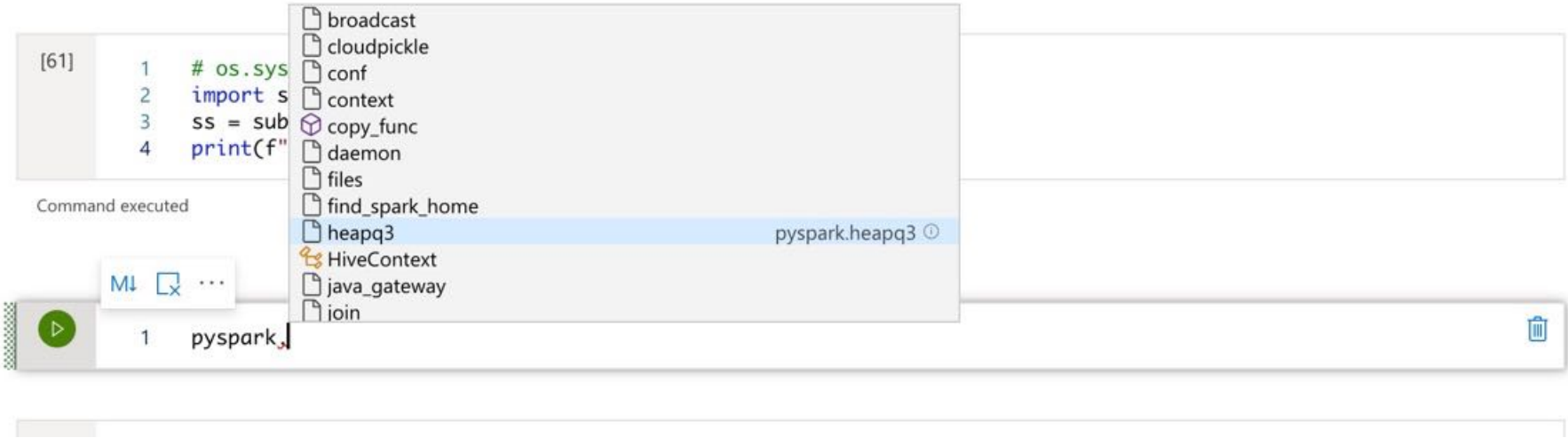
Notebook "Lev1" is running



Second notebook can't re-use the same cluster



PySpark Notebook – like Jupyter Notebook



- similar to Jupyter notebook, provides inline help
- can use regular python (current version 3.6.9)
- can use azureml libraries, sklearn libraries, matplotlib, seaborn, etc.
- can use spark/pyspark and MLlib libraries - https://spark.apache.org/docs/latest/

Synapse - Always Clusters = Pools
(for parallel processing)

SQL SQL SQL

SQL Engine
SQL Pools
(Dedicated & Serveless)

Spark Spark Spark

Spark Engine
Spark Pools
PySpark

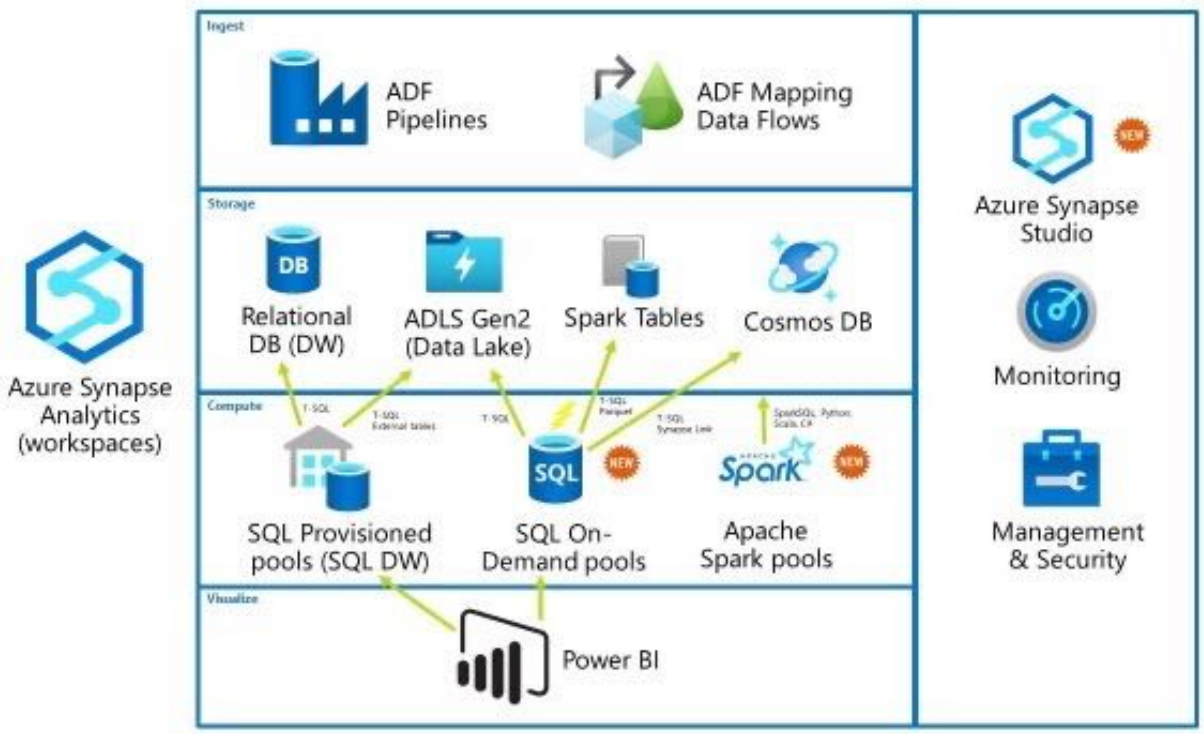
Note: Databricks can also be used as part of Synapse Integrate pipelines, although it is not part of Synapse Studio

Synapse Spark vs DataBricks

	Databricks	**Synapse Spark**
Spark	• very fast Runtime • can use GPU-clusters • standard/high-concurrency	• Open-source Apache Spark • support for .NET
Notebooks	• Databricks Notebooks • Real-time co-authoring • Automated versioning	• Nteract Notebooks • Has co-authoring of Notebooks, but one person needs to save the Notebook before another person sees the change. • Doesn't have automated versioning
Access Data Lake	• You need to mount a data lake before using it	• When creating Synapse, you can select your primary D.L.
Delta	• **Databricks Delta** has extra optimizations https://docs.microsoft.com/en-us/azure/databricks/delta/	• **Delta Lake** is open source https://docs.microsoft.com/en-us/azure/synapse-analytics/spark/apache-spark-what-is-delta-lake
Not the same	• Is not a data warehouse tool but rather a Spark-based notebook tool • Has a focus on **Spark**, **Delta Engine**, **MLflow** (open source platform to manage the ML lifecycle) and **MLR** (Machine Learning in R)	• Has both a traditional **SQL engine** as well as a **Spark engine** • Is a data warehouse (i.e. Synapse Analytics) + an interface tool (i.e. Synapse Studio)
Developer Experience	• Databricks UI, Databricks Connect (i.e. remote connect from Visual Studio Code, Pycharm, etc.) and soon Jupyter & RStudio UI within Databricks	• Spark-development only through Synapse Studio (not through local IDEs) • Doesn't have Git yet integrated within the Synapse Studio Notebooks

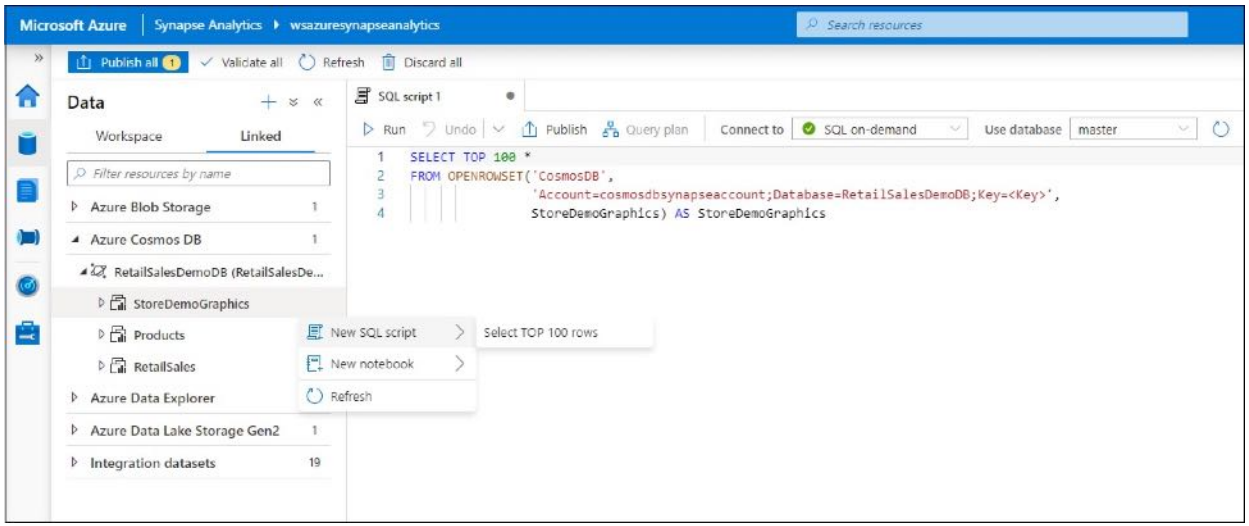
Synapse Pools

- Spark Pools
- SQL Pools:
  .. Dedicated SQL Pool (formerly SQL DW)
  .. Serverless Pool (on-demand, use SQL to query files, databases, or Spark Tables



Synapse SQL

Query from Data Source



Query from Notebook using pyspark or SQL

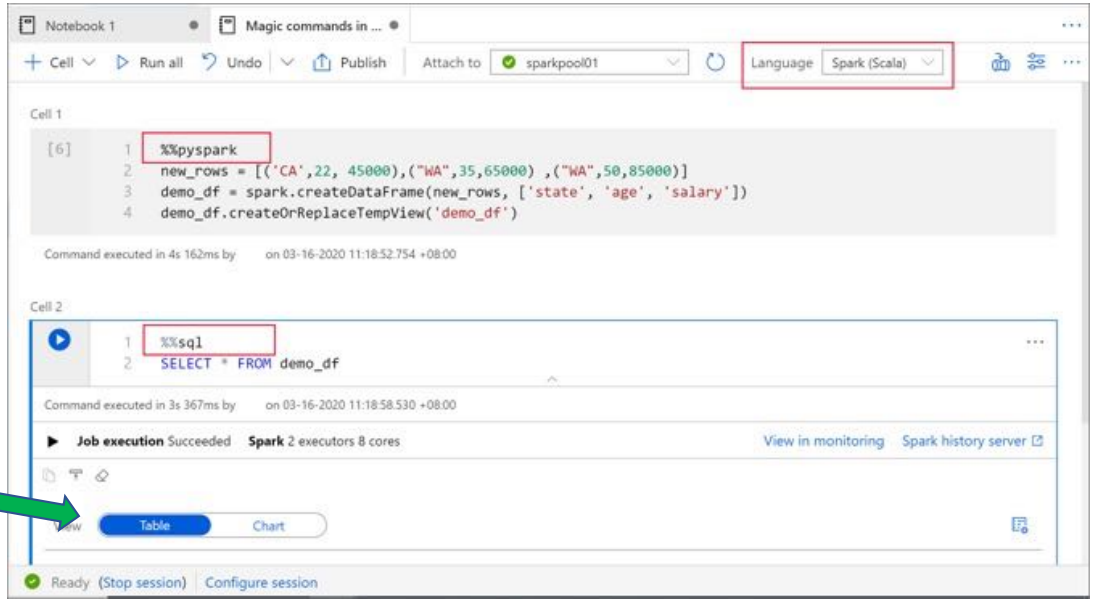
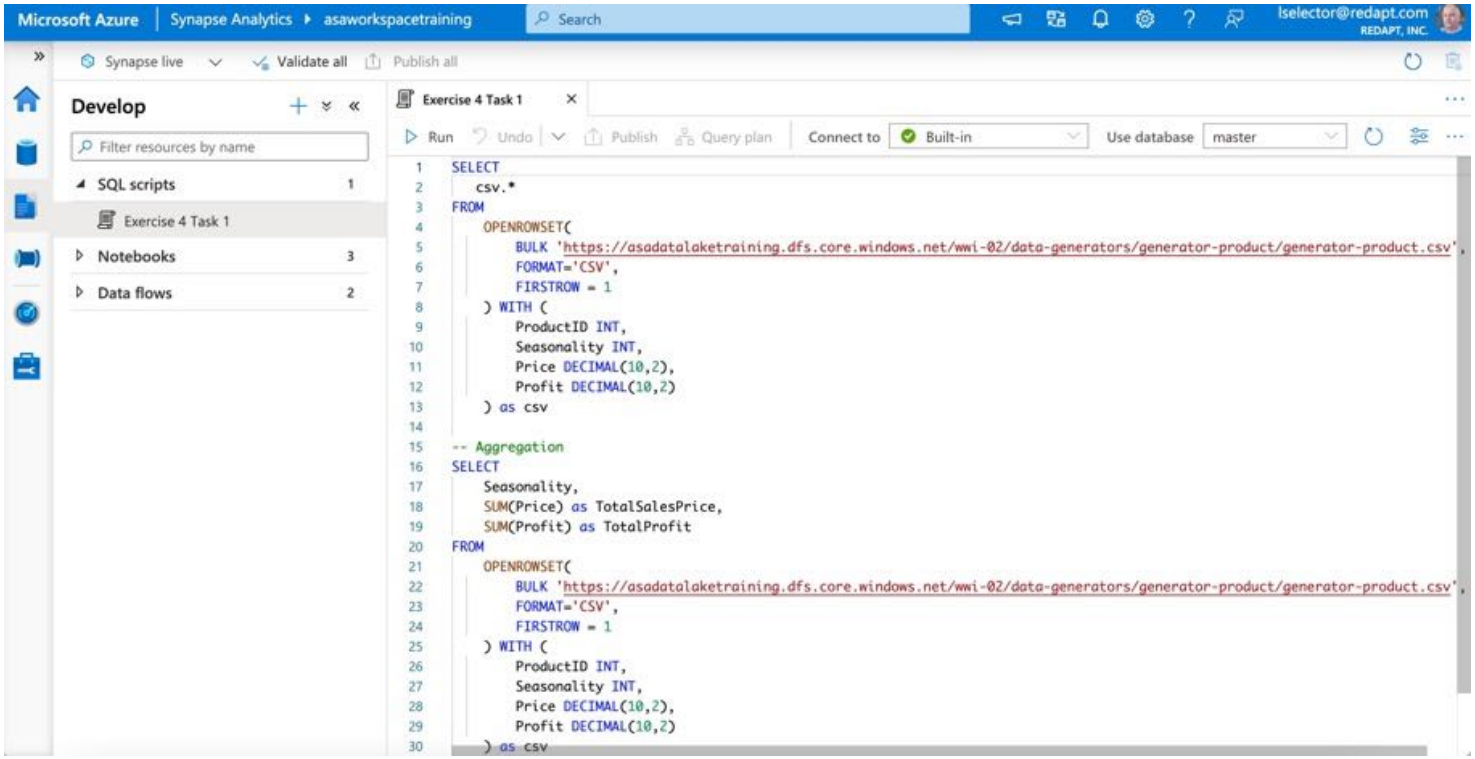


Table or Chart Output

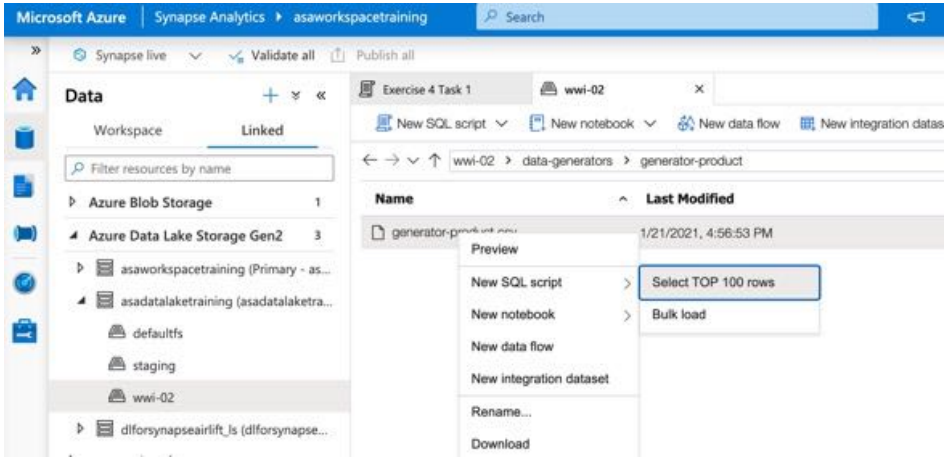
Synapse Notebooks:
https://docs.microsoft.com/en-us/azure/synapse-analytics/spark/apache-spark-development-using-notebooks?tabs=classical

Synapse SQL Scripts

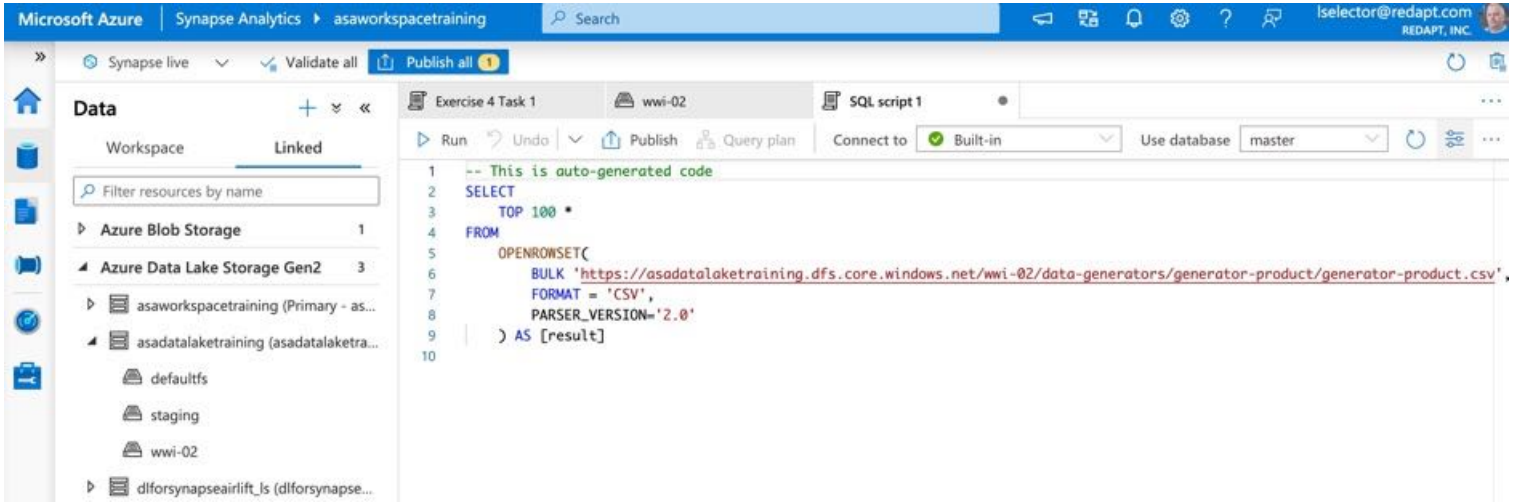
Example of SQL Query
from CSV files in Blob Storage



Synapse – getting link to a file in Blob Storage for SQL



- Locate the file in Blob storage
- Right-click on it and select **"Select top 100 rows"**
- You will get an SQL you can copy and use



Synapse Integrate

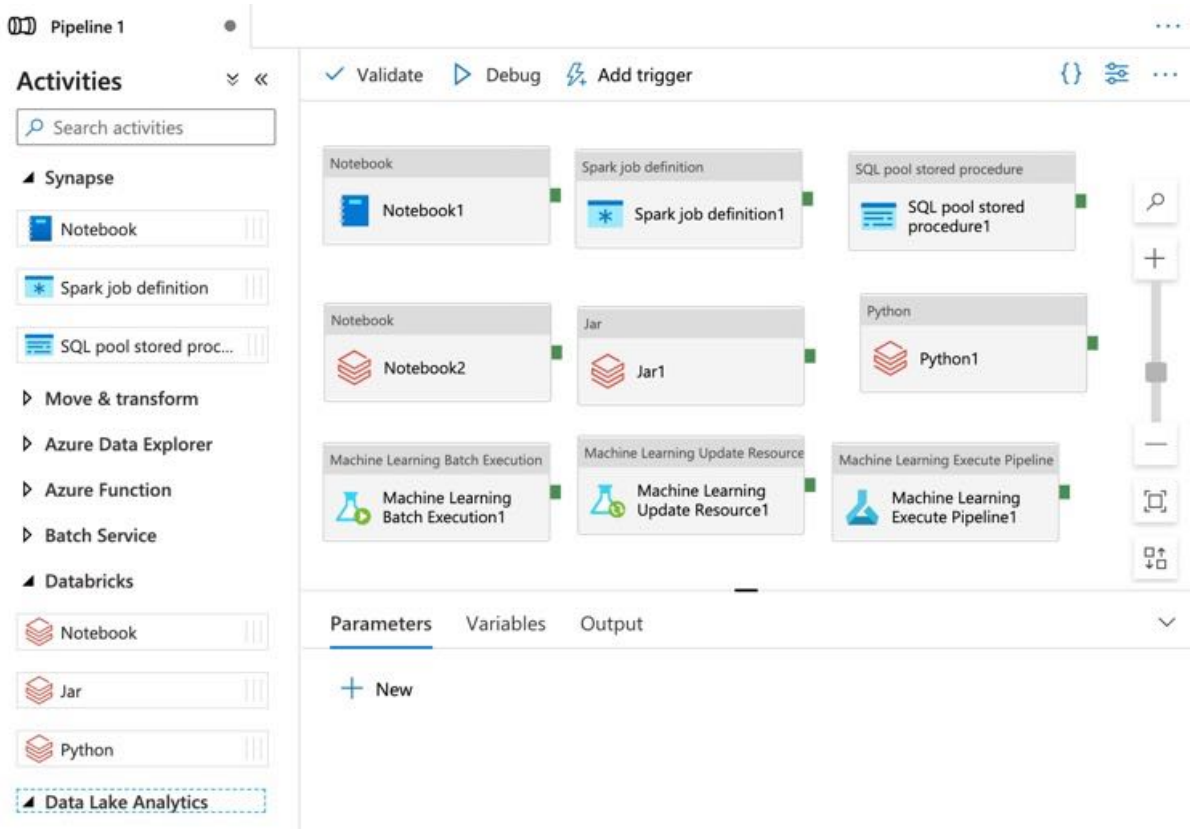
Creating Data Pipelines
Move and Process data

Examples:

Synapse Notebook
Synapse Spark job definition
Synapse SQL pool stored procedure

Databricks Notebook
Databricks Jar
Databricks Python

Machine Learning Batch Execution
Machine Learning Update Resource
Machine Learning Execute Pipeline



- https://docs.microsoft.com/en-us/azure/data-factory/
- https://docs.microsoft.com/en-us/azure/data-factory/transform-data-databricks-notebook
- https://docs.microsoft.com/en-us/azure/data-factory/transform-data-databricks-jar
- https://docs.microsoft.com/en-us/azure/data-factory/transform-data-databricks-python
- https://docs.microsoft.com/en-us/azure/data-factory/transform-data-using-machine-learning
- https://docs.microsoft.com/en-us/azure/data-factory/update-machine-learning-models
- https://docs.microsoft.com/en-us/azure/data-factory/transform-data-machine-learning-service

Azure Synapse Pricing and Docs

Azure Synapse Pricing
https://azure.microsoft.com/en-us/pricing/details/synapse-analytics/

Azure Synapse Documentation:
https://docs.microsoft.com/en-us/azure/synapse-analytics/

Good one-page tutorial on using pyspark notebook:
https://docs.microsoft.com/en-us/azure/synapse-analytics/spark/apache-spark-development-using-notebooks

Sample notebooks:
https://github.com/Azure-Samples/Synapse/tree/main/Notebooks

Visualize data
https://docs.microsoft.com/en-us/azure/synapse-analytics/spark/apache-spark-data-visualization-tutorial

Apache Spark:
https://en.wikipedia.org/wiki/Apache_Spark

Accessing files in Azure Gen2 storage:
https://github.com/Azure-Samples/Synapse/blob/main/Notebooks/PySpark/01%20Read%20and%20write%20data%20from%20Azure%20Data%20Lake%20Storage%20Gen2.ipynb

Azure Synapse Analytics and AI

Tutorial and Lab (step-by-step)
- https://github.com/microsoft/MCW-Azure-Synapse-Analytics-and-AI
- https://github.com/microsoft/MCW-Azure-Synapse-Analytics-and-AI/blob/master/Hands-on%20lab/HOL%20step-by%20step%20-%20Azure%20Synapse%20Analytics%20and%20AI.md

Note: MSFT provides this as Free training – you need to get credentials from them,
log-in to the Azure portal https://portal.azure.com/ with provided credentials.

Note - use incognito mode in Chrome, allow all cookies, stop all ad-blockers.

Notes: Three similar serverless solutions:
- Azure Serverless,
- Google BigQuery
- AWS Athena

Best practices for serverless SQL pool:
- https://docs.microsoft.com/en-us/azure/synapse-analytics/sql/best-practices-sql-on-demand

Capacity limits for dedicated SQL pool in Azure Synapse Analytics:
- https://docs.microsoft.com/en-us/azure/synapse-analytics/sql-data-warehouse/sql-data-warehouse-service-capacity-limits

REST APIs for dedicated SQL pool (formerly SQL DW) in Azure Synapse Analytics:
- https://docs.microsoft.com/en-us/azure/synapse-analytics/sql-data-warehouse/sql-data-warehouse-manage-compute-rest-api

Azure Machine Learning Tutorials:
- https://github.com/Azure/MachineLearningNotebooks/tree/master/tutorials -

Running Models from SQL or Power BI

Trained Model can be registered in SQL as a function !

Recommended approach:

1. **pickle file** of the model,
2. convert to **ONNX** (onnx.ai - Open Neural Network Exchange)
3. use it from SQL server:
   - https://docs.microsoft.com/en-us/azure/synapse-analytics/sql-data-warehouse/sql-data-warehouse-predict

Alternatively:
create a **stored procedure** to get the data,
and then **invoke model via cursor**

Also the model may be invoked via **REST API endpoint**.

Invoke model from PowerBI:
- https://github.com/alex-nogue/Machine_Learning_in_Power_BI#invoke-a-model-in-power-bi-using-azure-ml-service

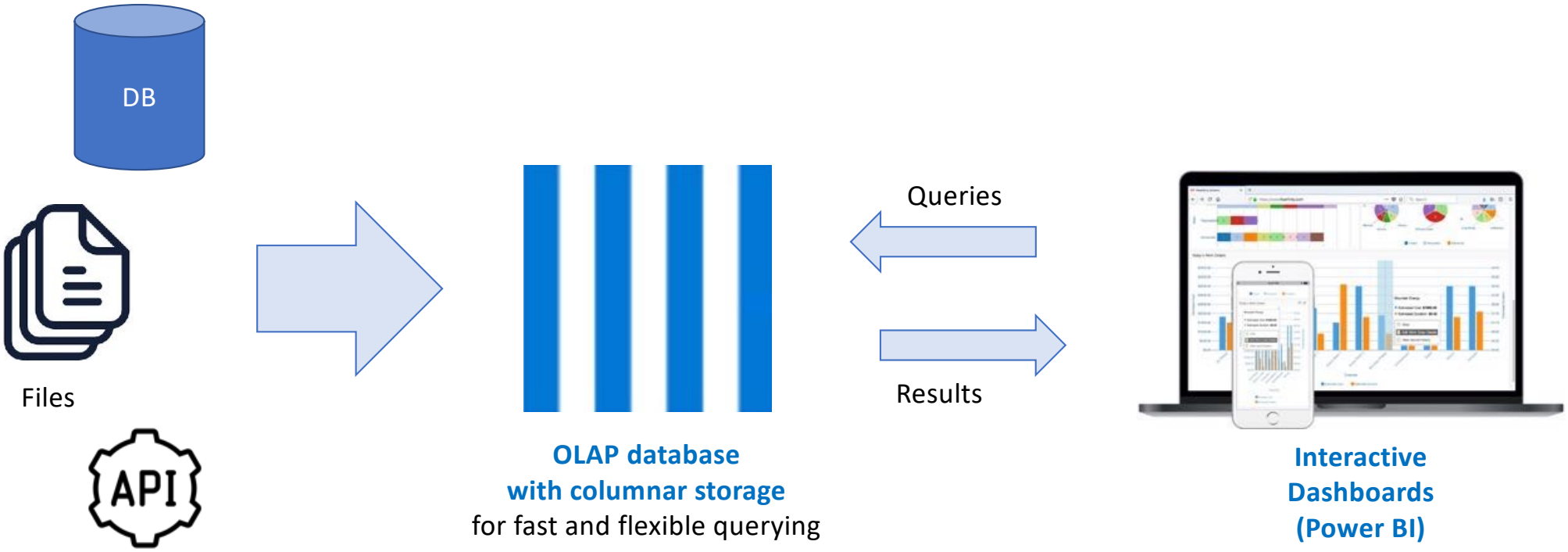
Three Ways to Run ML on SQL Server

- ML Service
- native PREDICT
- Remote Compute Context

- https://tsmatz.wordpress.com/2018/03/02/patterns-sql-server-machine-learning-services-r-and-python/

```
-----------------------------------------------------------
-- Sample table schema for storing a model and related data

CREATE TABLE [dbo].[Models]
(
    [Id] [int] IDENTITY(1,1) NOT NULL,
    [Model] [varbinary](max) NULL,
    [Description] [varchar](200) NULL
)
WITH
(
    DISTRIBUTION = ROUND_ROBIN,
    HEAP
)
GO

-----------------------------------------------------------
-- Copy command to load hexadecimal string of the model
-- from Azure Data Lake storage location

COPY INTO [Models] (Model)
FROM '<enter your storage location>'
WITH (
    FILE_TYPE = 'CSV',
    CREDENTIAL=(IDENTITY='access-id', SECRET='storage-key')
)

-----------------------------------------------------------
-- Query for ML predictions

SELECT d.*, p.Score
FROM PREDICT(
   MODEL = (SELECT Model FROM Models WHERE Id = 1),
   DATA = dbo.mytable AS d,
   RUNTIME = ONNX)
WITH (Score float) AS p;
```

Typical Data Organization for Analytics

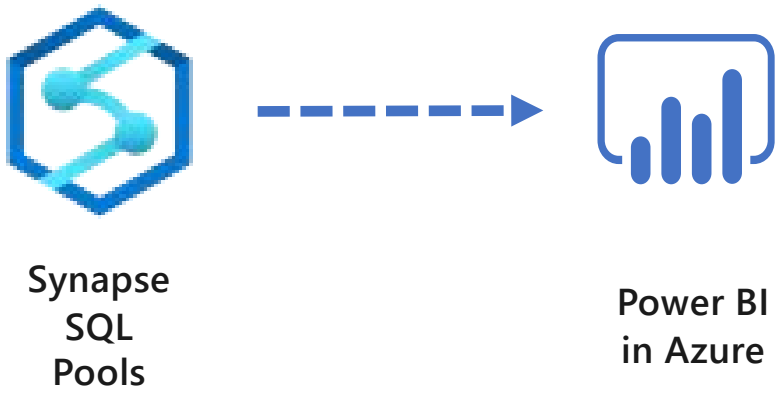


OLAP database
with columnar storage
for fast and flexible querying
(a.k.a. Analytical Store).

Interactive
Dashboards
(Power BI)

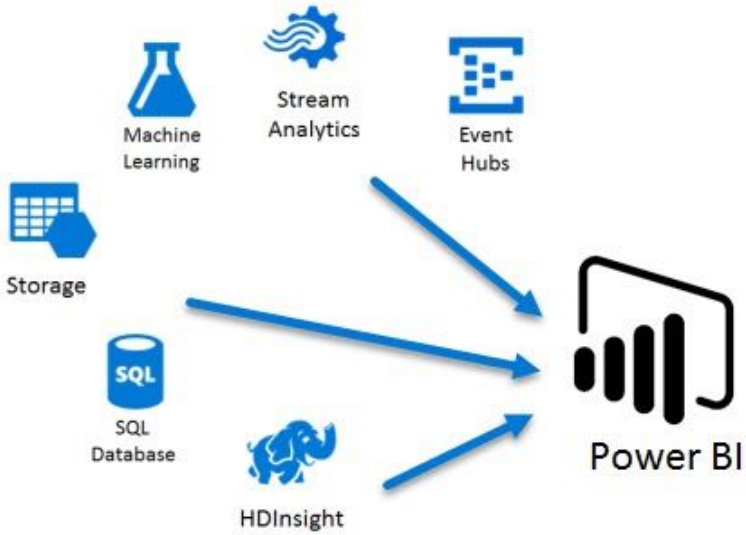
Rise and Fall of the OLAP Cubes:
https://www.holistics.io/blog/the-rise-and-fall-of-the-olap-cube/

Synapse – Simplified Architecture for Analytics (no intermediate analytical store)



Microsoft Power BI on Azure

- https://azure.microsoft.com/en-us/services/developer-tools/power-bi/
- https://powerbi.microsoft.com/en-us/

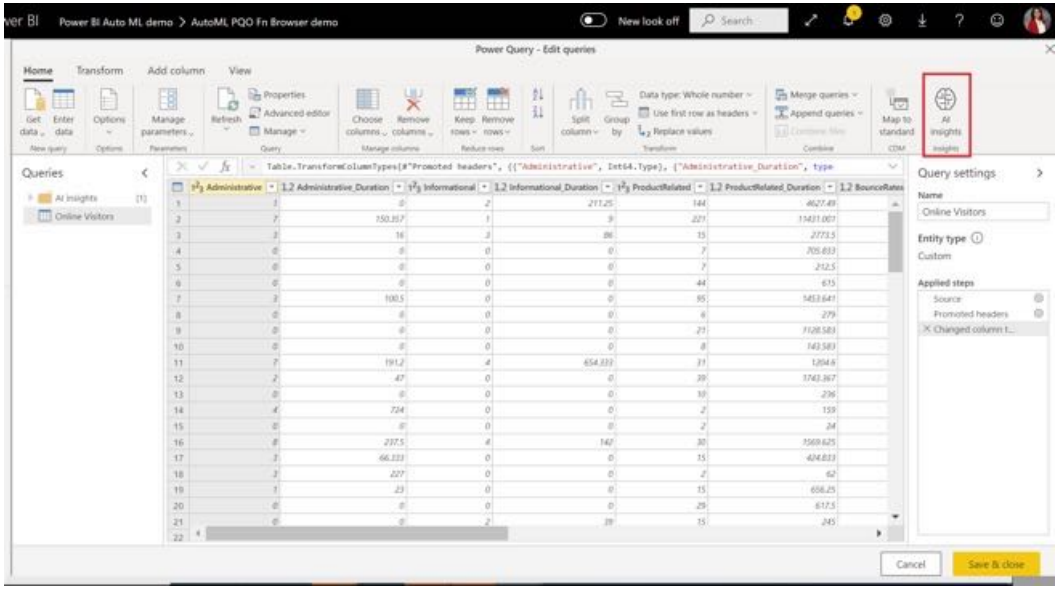


Power BI

No-code ML in Power BI is similar to AutoML in Azure ML Studio

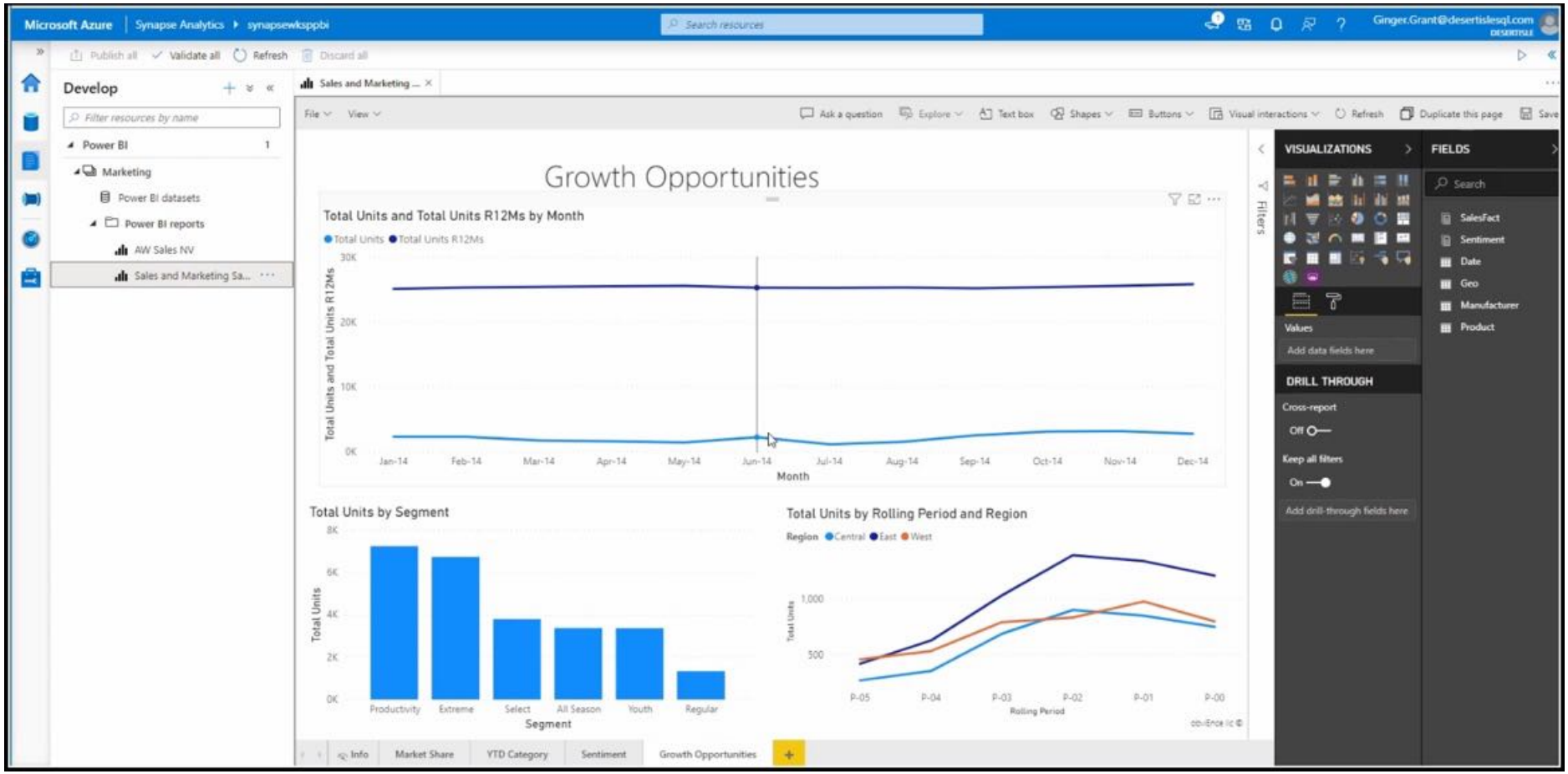
Tutorial: Build a Machine Learning model in Power BI
- https://docs.microsoft.com/en-us/power-bi/connect-data/service-tutorial-build-machine-learning-model

- Create a dataflow with the input data
- Create and train a machine learning model
- Review the model validation report
- Apply the model to a dataflow entity
- Using the scored output from the model in a Power BI report



Linking to Power BI from within Azure Synapse

Tutorial: https://docs.microsoft.com/en-us/azure/synapse-analytics/get-started-visualize-power-bi



Cosmos DB – Synapse Integration

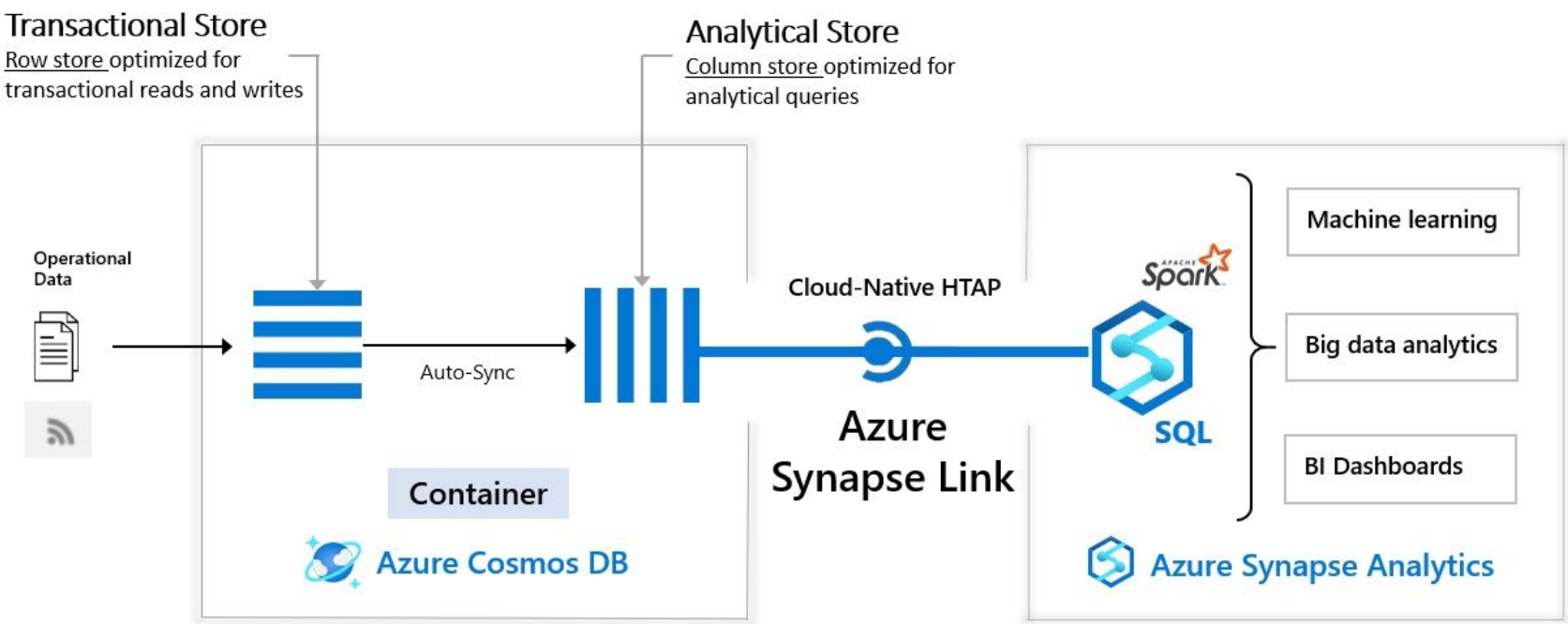


Diagram from https://docs.microsoft.com/en-us/azure/cosmos-db/synapse-link

More Links:

Visual Studio Code – good editor:
- https://code.visualstudio.com/

Download Integration Runtime:
- https://www.microsoft.com/en-us/download/details.aspx?id=39717 -

Download AZCopy:
- https://docs.microsoft.com/en-us/azure/storage/common/storage-use-azcopy-v10 -

AZCopy Documentation:
- https://docs.microsoft.com/en-us/azure/storage/common/storage-ref-azcopy-copy -

Transfer S3 using AzCopy:
- https://docs.microsoft.com/en-us/azure/storage/common/storage-use-azcopy-s3 -

Differences between Standard and Premium storage:
- https://www.danielstocker.net/what-is-the-difference-between-azure-standard-storage-and-azure-premium-storage/ -
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